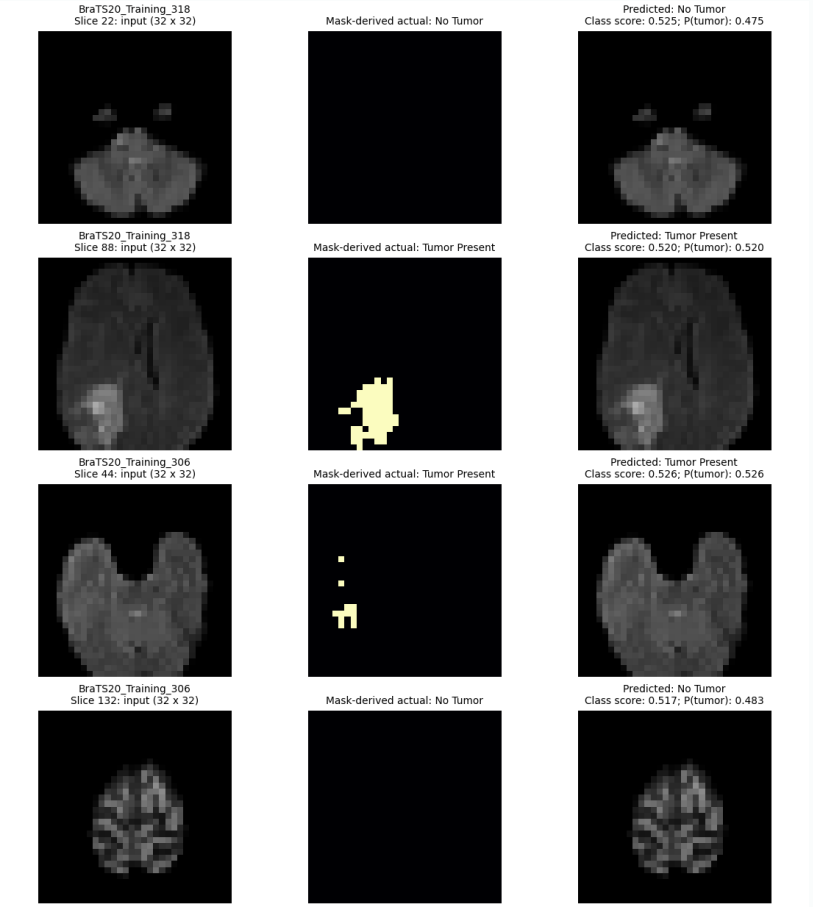
Brain Tumor Detection Using MRI Images

Brain + MRI Research Platform | Academic project presentation

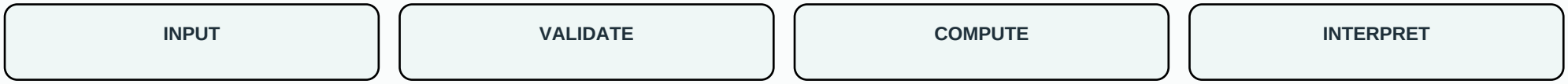
- Evidence-led biomedical imaging, tutorial delivery and experimental AI workflow
- Research-only: no clinical diagnosis or patient-level claim



Presentation evidence audit

Revision checklist applied to the earlier sparse 33-slide deck.

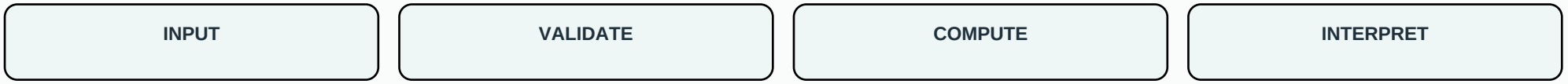
- Expanded to technical diagrams, tables and source-backed metrics.
- Embedded saved plots, original notebook outputs and project visualizations.
- Expanded CNN to eleven methodology/evidence slides.
- Covered T1–T15 with aim, inputs, processing, output and limitation.



Problem definition

MRI analysis requires reproducible processing before output can be interpreted.

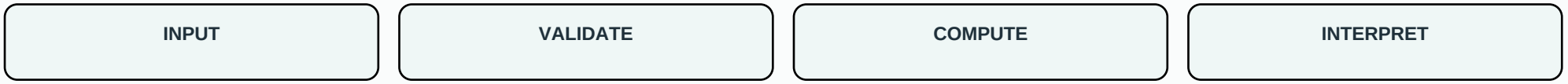
- MRI volumes may include invalid values, intensity variation and absent modalities.
- Supplied segmentation is reference ground truth—not a model prediction.
- Small local data supports teaching/pipeline validation, not clinical validation.



Platform architecture

Independent browser modules reuse validated MRI and analytics utilities.

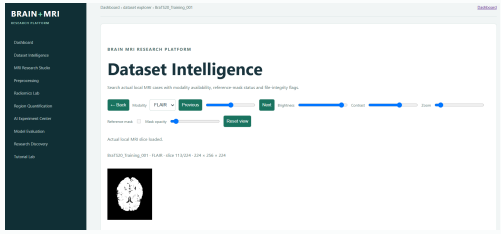
- Browser: Dashboard, Tutorial Lab, Image Analysis Studio, MRI Research Studio.
- FastAPI: validation, data-URL image serialization, case discovery and reporting.
- Core: NIfTI I/O, preprocessing, SIFT/RANSAC, analytics and CNN inference.

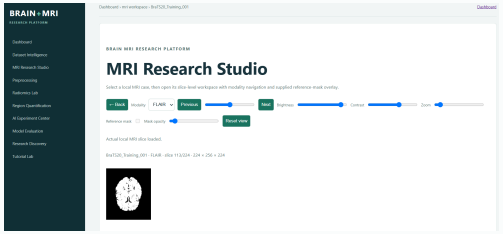


Frontend modules: contextual research workspace

Actual application screens show Dataset Intelligence and MRI Research Studio.

- Dataset Intelligence provides local case selection, modality context and actual slice preview.
- MRI Research Studio separates workflow selection from case-specific analysis and upload inference.
- The visible Back action follows contextual module hierarchy rather than always returning to Dashboard.

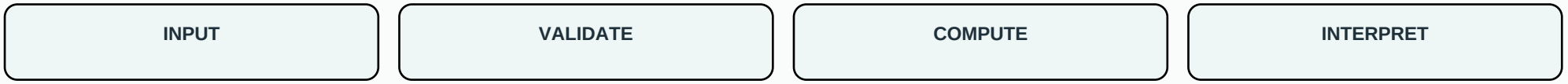




Data lineage and safeguards

Every visible result has a traceable source and status.

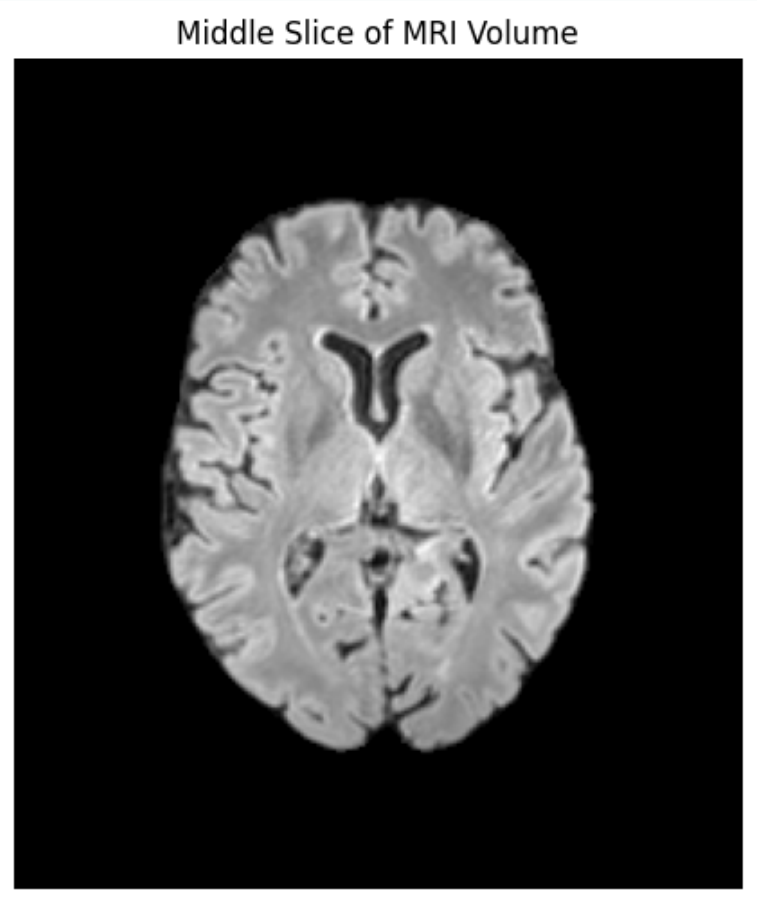
- Local FLAIR/SEG → validate → preprocess → algorithm/checkpoint → PNG/metrics → UI.
- Missing masks stay missing; they are never converted to negative labels.
- The persistent checkpoint is reused; request handling never trains a new model.



Dataset audit: local BraTS2020 evidence

Actual findings from CNN_DATASET_AUDIT.md.

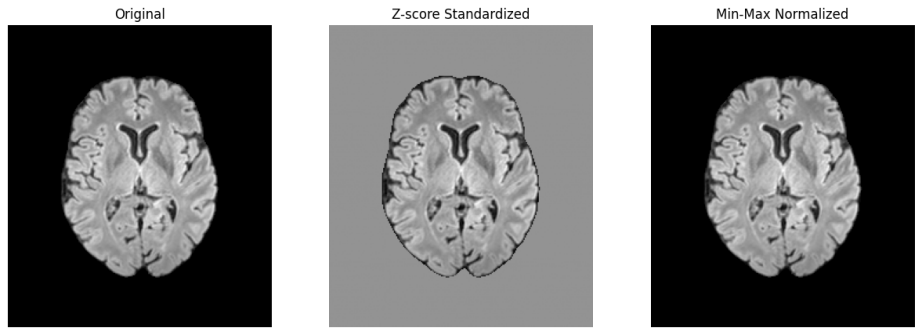
- 8 case folders; FLAIR 8/8; SEG 6/8. Two unsegmented cases are excluded from supervision.
- Six FLAIR/SEG pairs are 240 × 240 × 155. Labels 0,1,2,3; y=1 iff mask > 0.
- All eligible labelled slices: 798 (355 negative, 443 positive).



MRI preprocessing contract

The exact branch is shared by training, validation, test and upload inference.

- Clean NaN/Inf → 160 × 160 centre crop (when applicable) → per-slice min–max.
- $x' = (x - \min(x)) / (\max(x) - \min(x))$; resize to 128 × 128; one FLAIR channel.
- This contract is implemented in her_core/cnn.py, not recreated per endpoint.

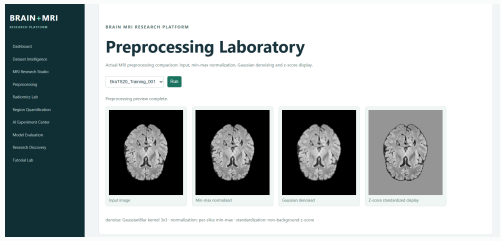


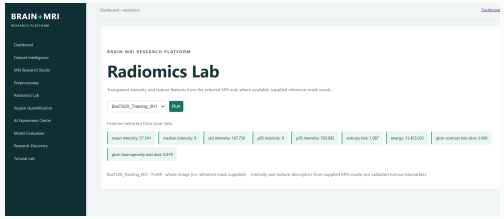
Research-only · not a clinical diagnosis

Implemented application analysis modules

Frontend outputs demonstrate that preprocessing and radiomics are separate functional modules.

- Preprocessing Laboratory renders original, cleaned, normalized and contrast-enhanced MRI stages.
- Radiomics Lab selects a case/modality and exposes feature-analysis actions backed by actual local data.
- These modules are analysis tools; they do not report clinical diagnoses.

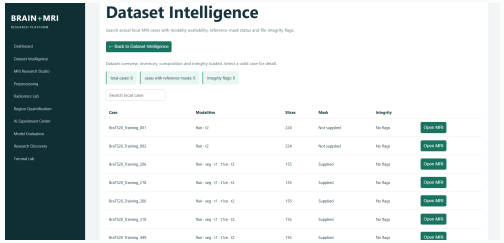


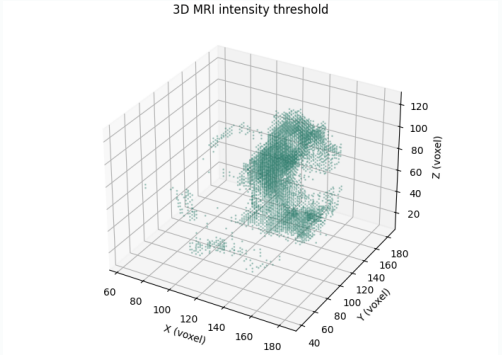


MRI Workspace and Dataset Explorer

Application modules render data rather than raw JSON.

- Case/modality/slice controls call the validated slice endpoint and return compact PNG data URLs.
- Brightness, contrast, zoom and reference-mask overlay apply only where the actual input exists.
- Dataset Explorer displays case inventory, modality availability, slice counts and integrity state.

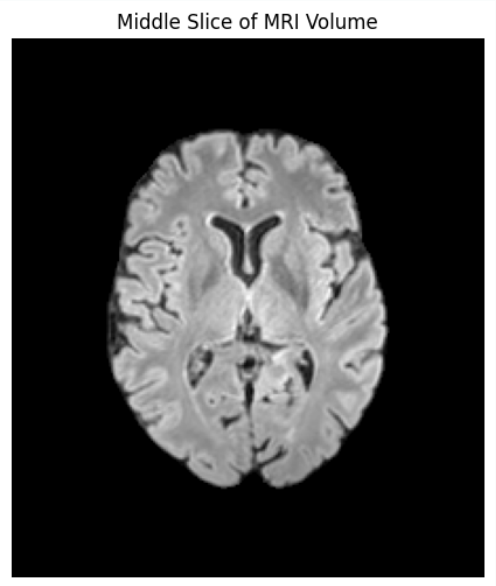


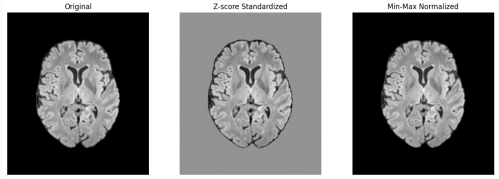


T1–T3: inspection, preprocessing and data quality

Foundational workflows establish data conditions before analysis.

- T1: discover local cases/modalities and show a middle slice.
- T2: clean invalid values; compare z-score and min–max transformations.
- T3: report missing modalities/duplicates without destructive assumptions.

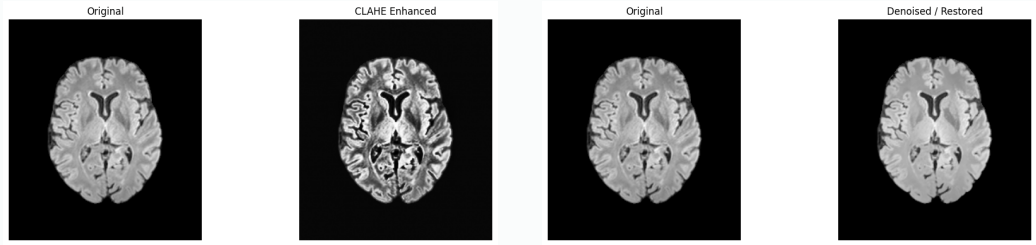




T4: biomedical image analysis

Aim: make enhancement and quality operations inspectable.

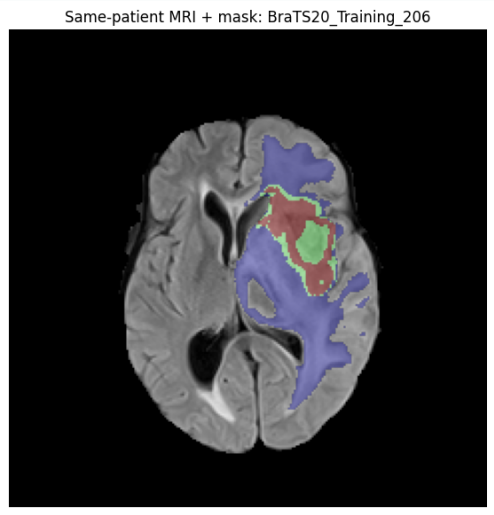
- Input: local MRI slice. CLAHE enhancement and bilateral denoising are displayed side by side.
- Interpretation: altered contrast is presentation support—not tumour segmentation/diagnosis.
- Implementation: notebook operations and reusable preprocessing helpers.

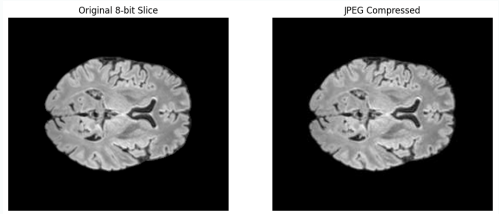


T4: reference overlay and compression

Reference masks remain distinct from processing effects.

- Same-case FLAIR + supplied SEG are overlaid as reference evidence.
- JPEG comparison demonstrates storage/artefact trade-offs for teaching.
- No predicted segmentation mask is claimed.

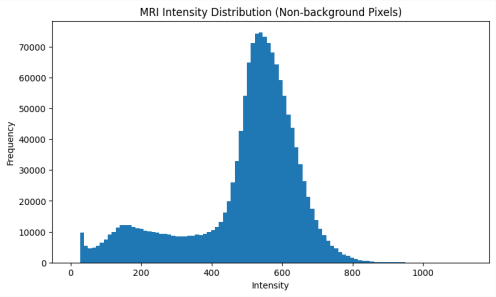


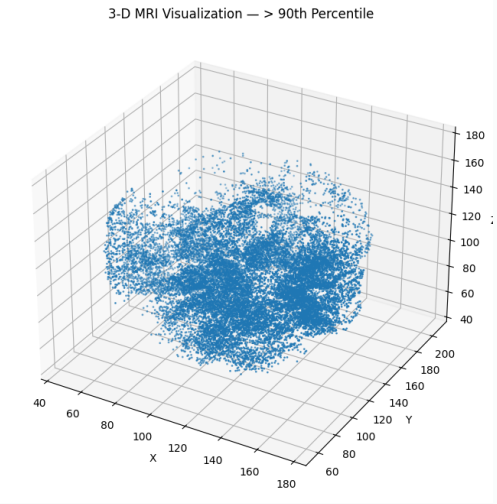


T4: intensity and 3-D context

Additional notebook evidence links slice-level work to volume context.

- Histogram uses non-background MRI intensity values, making intensity distribution explicit.
- 3-D threshold display is a visualisation of high-intensity voxels, not a tumour segmentation.
- These figures support inspection before feature extraction or classification.



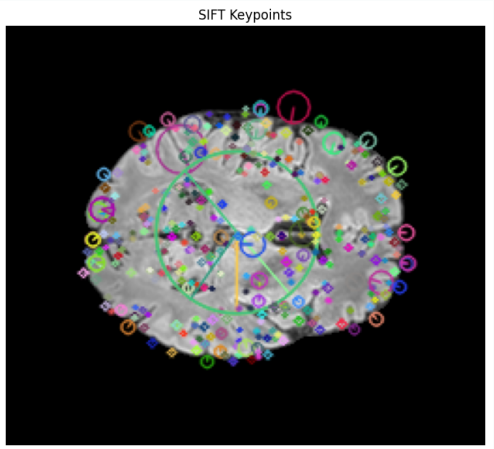


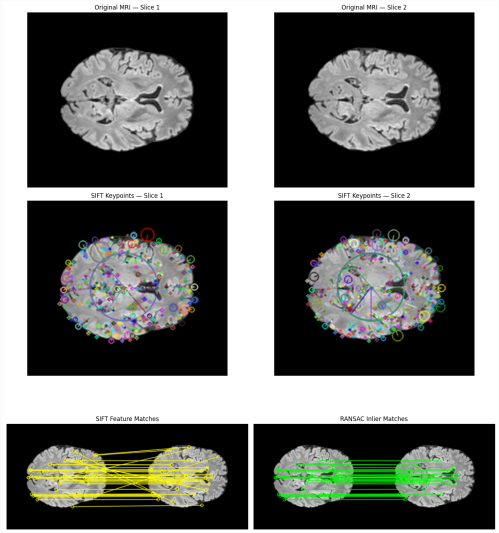
Research-only · not a clinical diagnosis

T5: SIFT + RANSAC

Aim: robust correspondence evidence between nearby MRI slices.

- SIFT detects scale-invariant keypoints and descriptors from normalized intensity.
- Descriptor matching proposes candidates; RANSAC estimates a robust transform/rejects outliers.
- Output supports image correspondence analysis, not tumour presence.

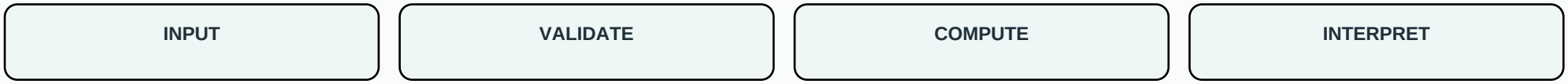




T5 CNN: target and leakage control

Binary slice labels derive only from reference-mask pixels.

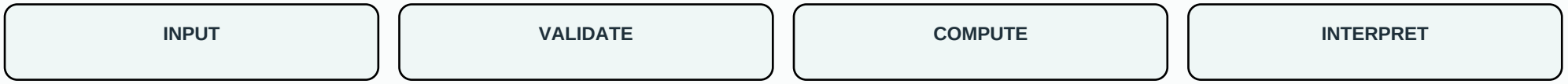
- y = 1 if any(mask > 0), otherwise 0; unlabelled cases are excluded.
- Case separation: train 4 cases/132 slices; validation 1/35; held-out test 1/35.
- No patient contributes slices to more than one partition.



T5 CNN: layer-by-layer architecture

Saved model brats-flair-slice-cnn-v1 | input 128 × 128 × 1.

- Conv2D(32,3×3,ReLU,same) + pool → Conv2D(64) + pool → Conv2D(128) + pool.
- Flatten → Dense(128, ReLU) → Dropout(0.5) → Dense(1, sigmoid).
- Sigmoid is model-estimated P(tumour slice); evaluation threshold = 0.50.



T5 CNN: experiment configuration

Persistent run from scripts/train_brain_tumor_cnn.py.

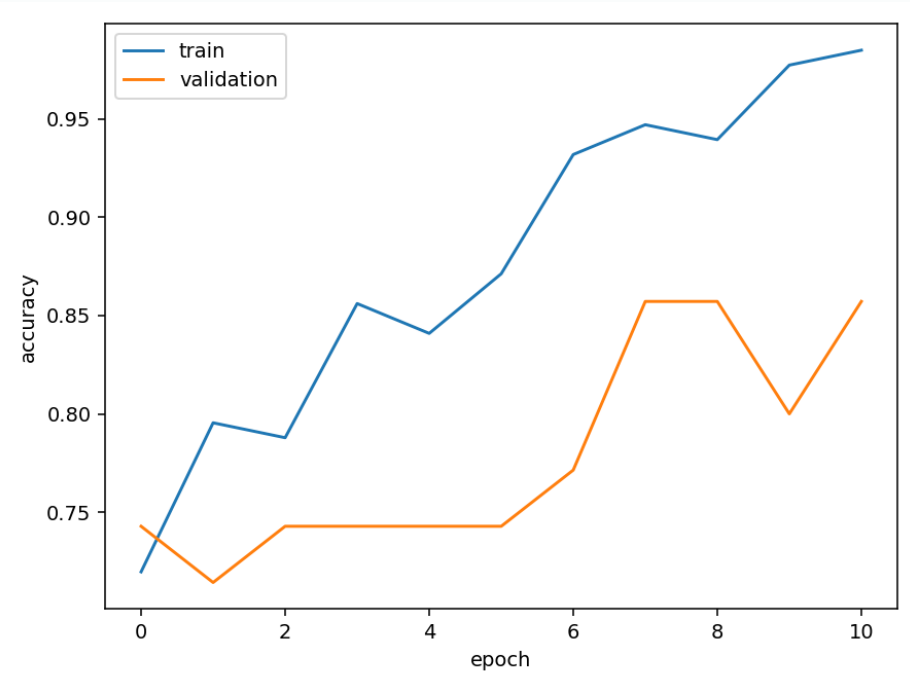
- Fixed seed 42; 12 requested epochs; best validation-loss checkpoint persisted.
- Training-only weights: no-tumour 1.1786; tumour 0.8684.
- E

Partition	Cases	Slices	Role
Train	457,449,218,206	132	Optimisation
Validation	306	35	Selection
Held-out test	318	35	Experimental test

T5 CNN: training and validation accuracy

Actual persisted history; 11 recorded epochs.

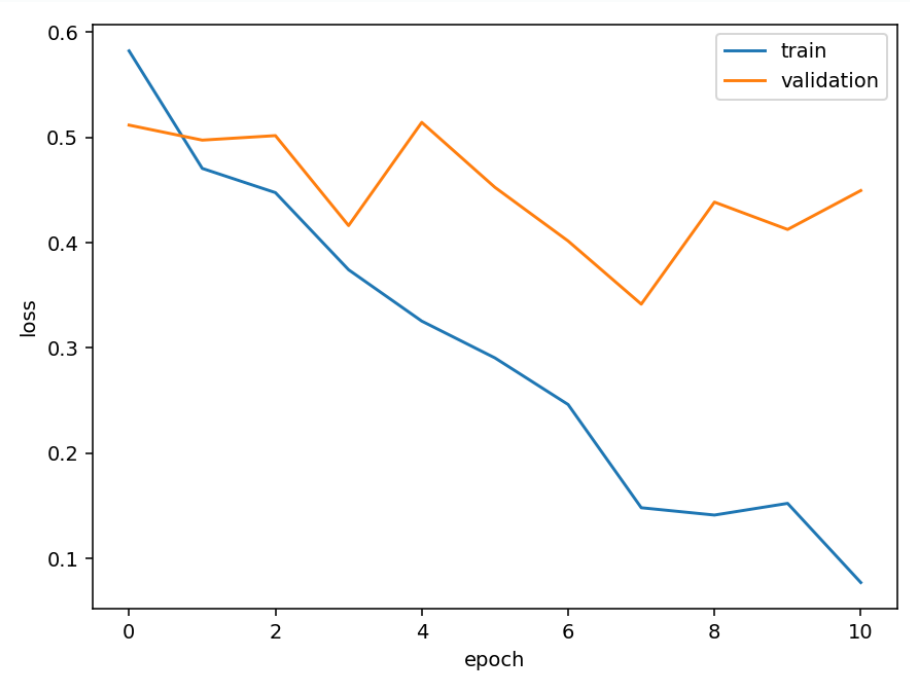
- Training accuracy: 0.7197 → 0.9848. Validation ends at 0.8571.
- The gap is a small-cohort caution, not deployment evidence.
- Validation is for model selection; held-out reporting is separate.



T5 CNN: training and validation loss

Actual persisted loss history; checkpoint selected by validation loss.

- Training loss: 0.5823 → 0.0772. Validation loss is non-monotonic and ends at 0.4496.
- The curves motivate independent patient-level evaluation; they do not prove generalisation.

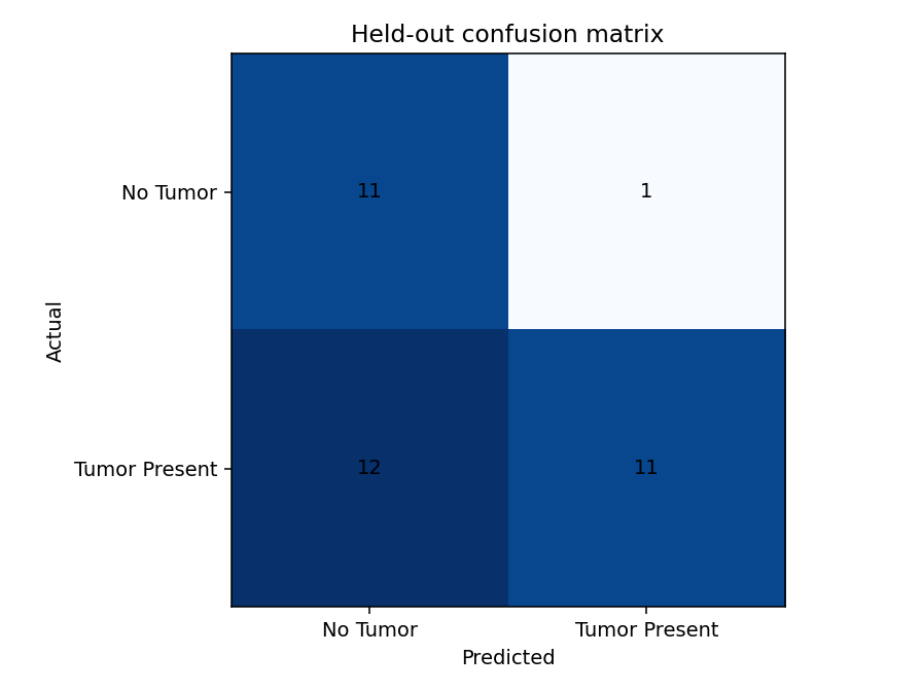


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T5 CNN: held-out confusion matrix

Actual one-patient held-out test: 35 slices.

- [[11,1],[12,11]] = 11 TN, 1 FP, 12 FN, 11 TP.
- Tumour recall = 0.4783: 12/23 positive slices missed at threshold 0.50.
- Negative slice result cannot rule out tumour.

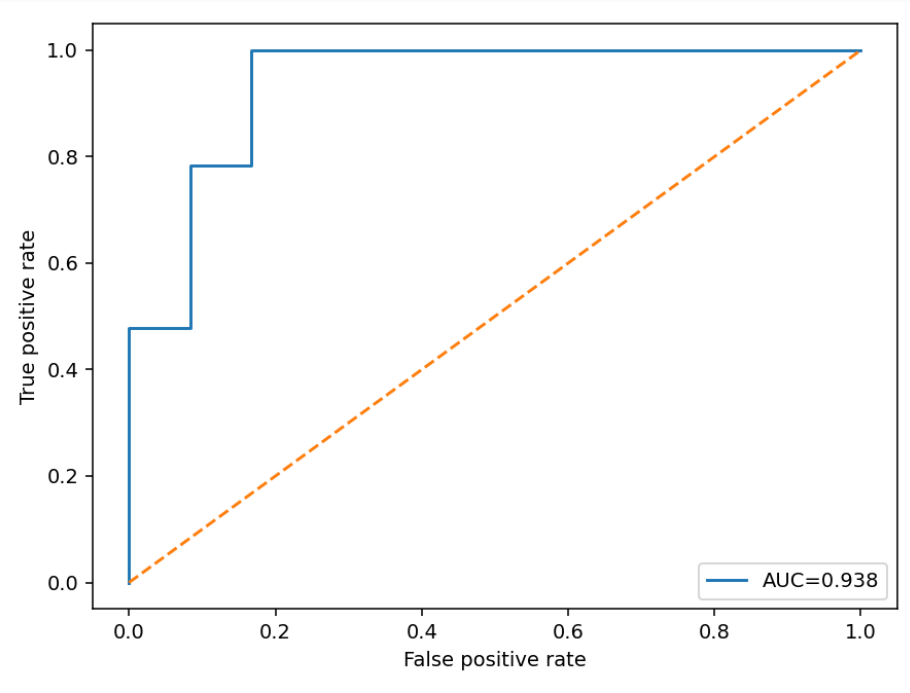


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T5 CNN: ROC and threshold interpretation

Score ranking and thresholded classification answer different questions.

- Held-out ROC-AUC = 0.9384; thresholded accuracy = 0.6286; tumour recall = 0.4783.
- High AUC can coexist with modest recall at one fixed threshold.
- One held-out patient means neither statistic establishes clinical calibration.



T5 CNN: classification report

Actual held-out metrics (35 slices).

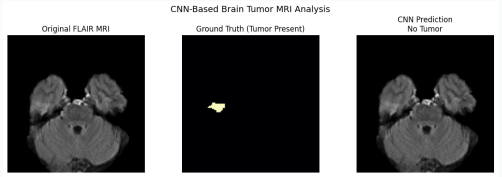
Class	Precision	Recall	F1	Support
No Tumor	0.4783	0.9167	0.6286	12
Tumor Present	0.9167	0.4783	0.6286	23
Macro average	0.6975	0.6975	0.6286	35
Weighted average	0.7664	0.6286	0.6286	35

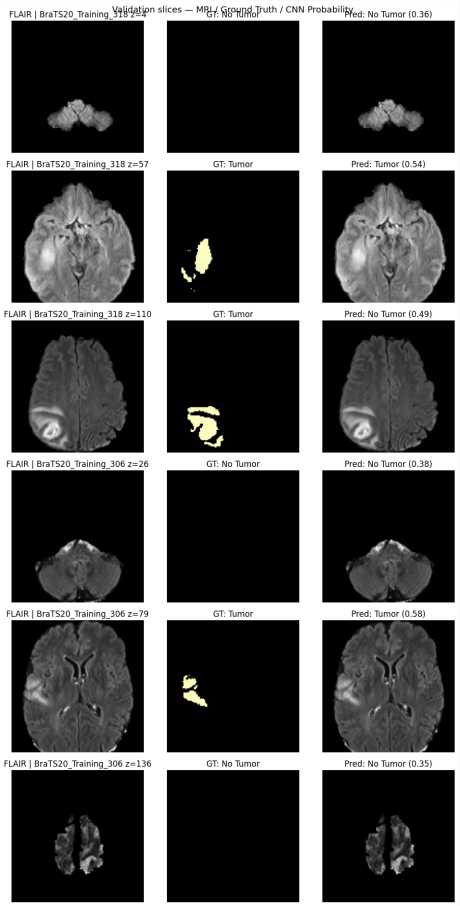
Research-only · not a clinical diagnosis

T5 CNN: actual inference visualization

Notebook evidence demonstrates genuine preprocessed-slice predictions.

- UI renders original MRI and exact 128 × 128 model input.
- Tutorial data can show actual label separately; arbitrary uploads do not have ground truth.
- Probabilities are model estimates, never diagnostic certainty.

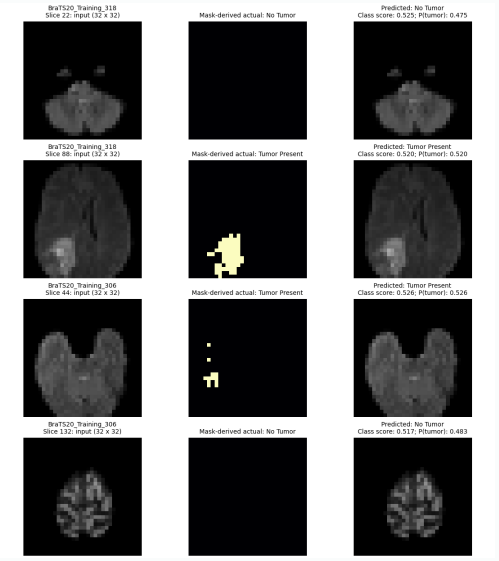


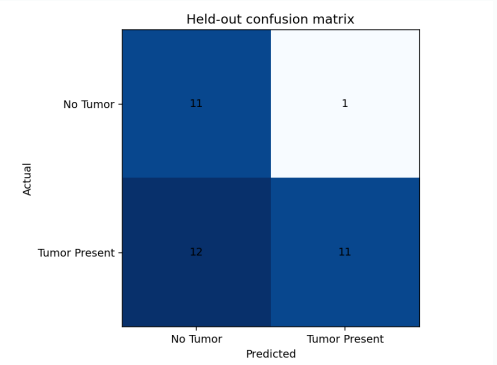


T5 CNN: result interpretation

Model output is technically useful only within its documented scope.

- A sigmoid probability ranks the uploaded/preprocessed slice under the trained experimental model.
- Thresholding at 0.50 creates a class label; it does not transform probability into clinical certainty.
- False negatives in the held-out matrix and one-patient test design require a conservative interpretation.

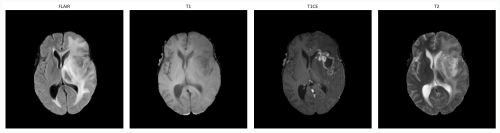


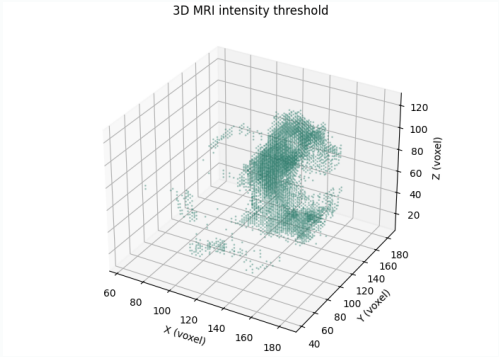


T6: multimodal MRI visualization

Aim: compare compatible modalities and supplied segmentation context.

- Same-case volumes are rendered with consistent orientation.
- SEG pane is supplied reference where present—not CNN output.
- Modality comparison precedes downstream interpretation.





T7: MRI review queue

Aim: operational case review from actual data availability.

- Input: discovered local cases plus FLAIR/SEG availability.
- Output: review state and completeness status; T7–T9 do not request synthetic text examples.
- Availability is operational metadata, not a medical finding.



T8: healthcare data analysis

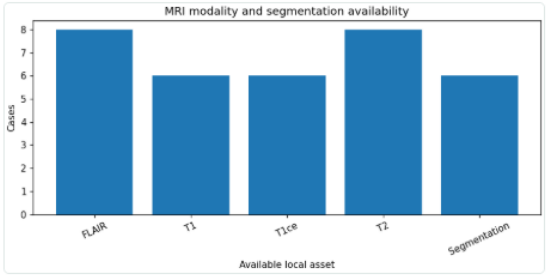
Aim: inspect dataset inventory and supplied tumour-burden information.

- Inputs: local metadata and SEG only where provided.
- Processing: counts, availability checks and volume-derived summaries.
- Small local summaries are repository descriptions—not epidemiological estimates.

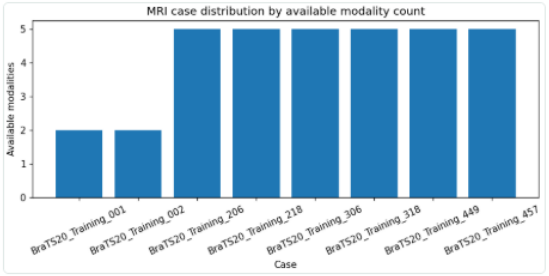
mri cases	cases with flair	cases with segmentation	cases without segmentation
8	8	6	2

Planning summaries describe the supplied MRI and annotation inventory. Segmentation presence is an annotation attribute, not a diagnosis for unsegmented cases.

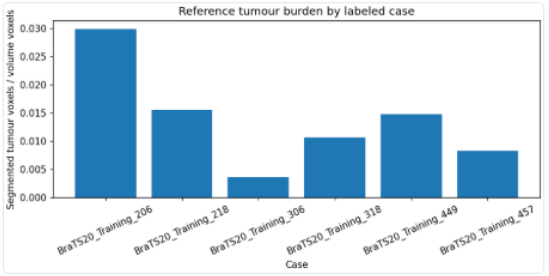
MRI modality and segmentation availability



MRI case distribution by available modality count



Reference tumour burden by labeled case

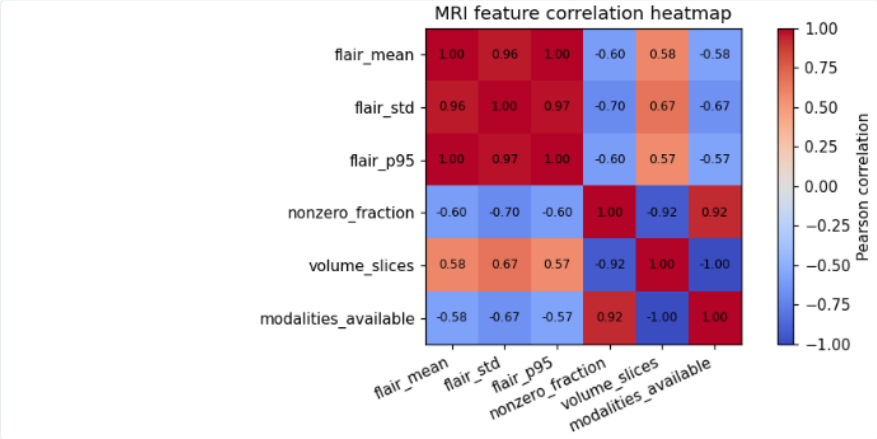


T9: feature exploration

Aim: derive MRI features for descriptive statistics.

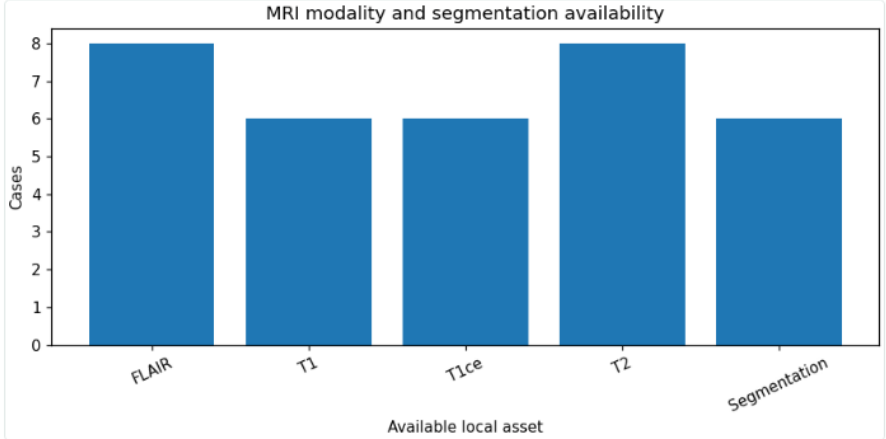
- Normalized images/volumes yield intensity and simple morphology-derived exploration.
- Associations are descriptive; local cohort size limits inference and generalisation.

MRI feature correlation heatmap



Pairwise correlations of MRI-derived features. Small case count prevents confirmatory inference.

MRI modality and segmentation availability

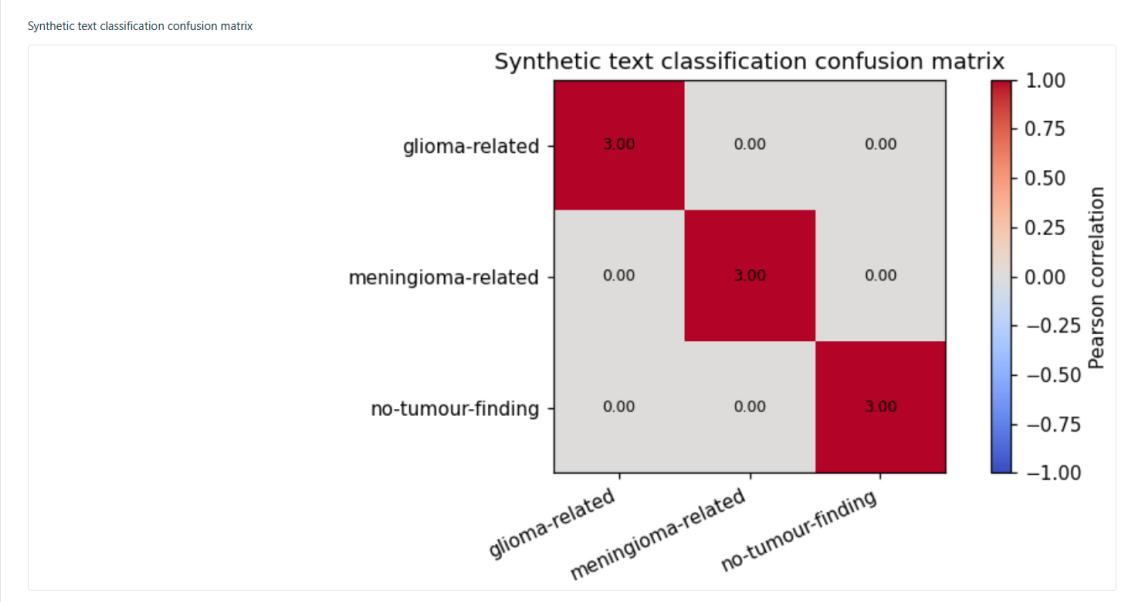


Availability is read from local case folders; missing assets are not imputed.

T10: text classification

A synthetic teaching-report workflow, distinct from MRI imaging.

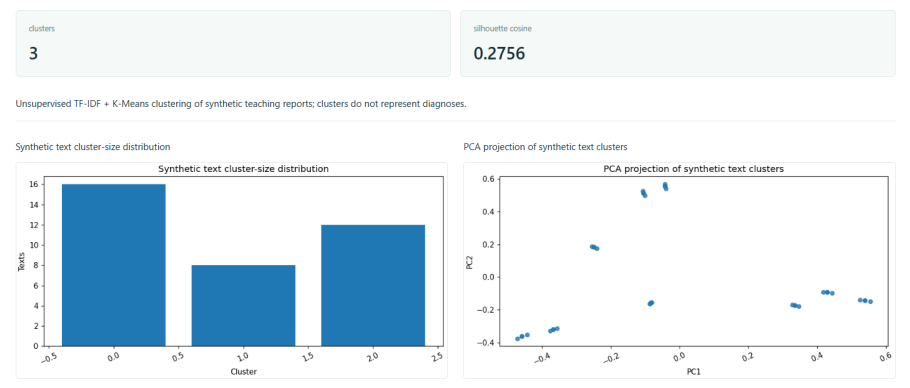
- Supported synthetic CSV examples are restricted to T10–T12.
- TF–IDF features feed logistic classification and class-oriented tutorial outputs.
- No real patient reports, clinical NLP accuracy or diagnosis is claimed.



T10: clustering

Clustering remains independent from text classification.

- TF–IDF vectors feed K-Means; cluster IDs describe lexical similarity, not diagnoses.
- Separate UI controls preserve the difference between supervised and unsupervised workflows.



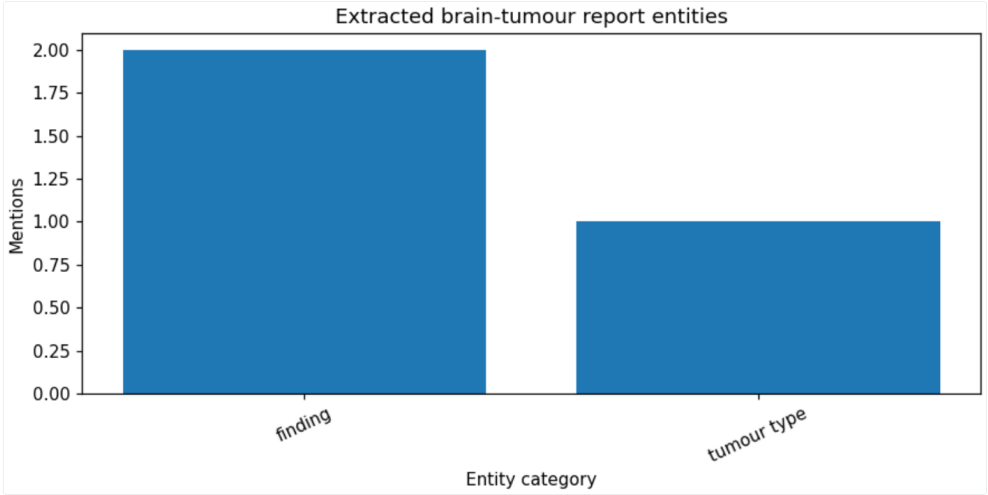
T11: entity extraction

Transparent lexicon and measurement-rule extraction.

- Synthetic text is processed with term and pattern matching.
- Output is structured teaching evidence, not a trained clinical NER system.

Extraction is rule-based and transparent, not clinical-grade NER. No annotated evaluation corpus is available, so accuracy is not calculated.

Extracted brain-tumour report entities



T12: ranking and disambiguation

TF–IDF cosine ranking of synthetic report passages.

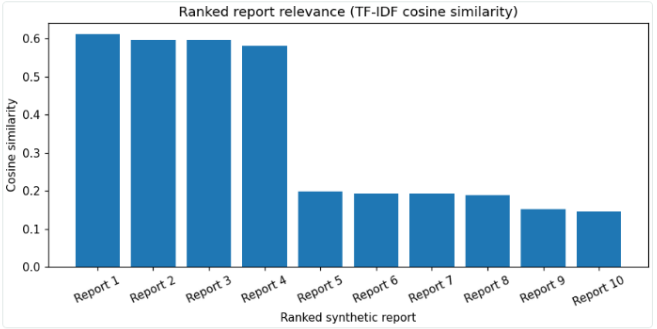
- Query/corpus → vectorisation → cosine similarity → ranked result with ambiguity context.
- This demonstrates retrieval mechanics, not clinical decision support.

top relevance

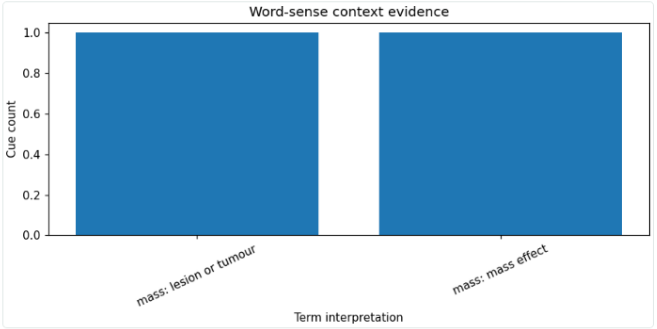
0.6109

Lexical relevance and context-cue counts are not diagnosis or clinical recommendation. Synthetic templates are marked as such.

Ranked report relevance (TF-IDF cosine similarity)



Word-sense context evidence



T13: PCA and t-SNE

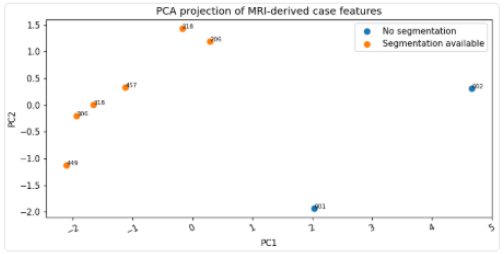
Aim: lower-dimensional exploratory projections of standardised MRI features.

- PCA reports linear projection/explained-variance evidence; t-SNE offers nonlinear neighbourhoods.
- Visual clusters are hypothesis-generating, not validated biological subtypes.

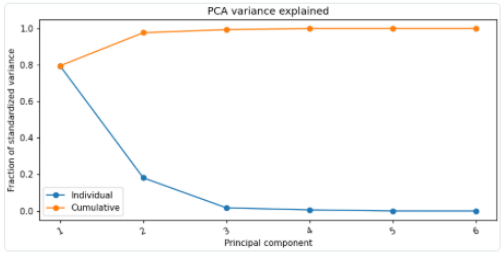
mri cases	cases with flair	cases with segmentation	cases without segmentation	first two variance
8	8	6	2	0.9771

PCA and t-SNE use standardized real MRI-derived features. Point color denotes segmentation availability, not a tumour class, because unsegmented cases have unknown reference status.

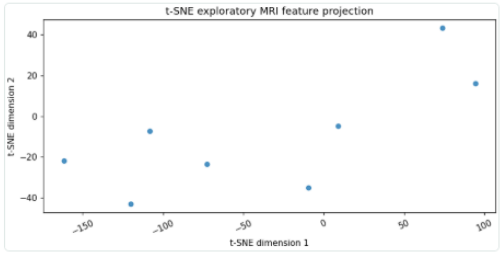
PCA projection of MRI-derived case features



PCA variance explained



t-SNE exploratory MRI feature projection



t-SNE is exploratory; distances and apparent clusters are unstable for this small local dataset.

T14: tumour-burden associations

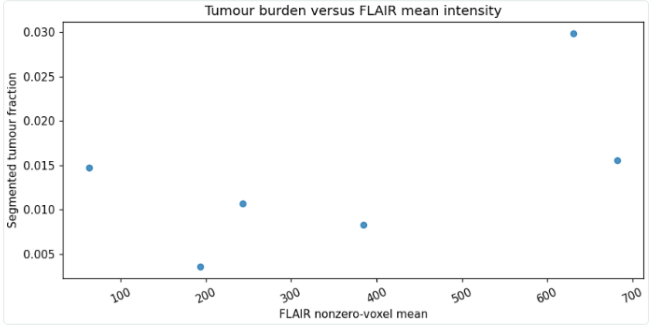
Aim: exploratory associations with supplied reference-mask burden.

- Feature summaries are paired with SEG-derived burden only when a reference mask exists.
- Association plots are non-causal and missing masks remain missing.

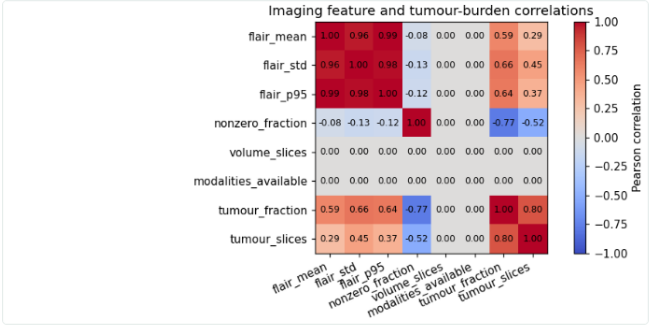
mri cases	cases with flair	cases with segmentation	cases without segmentation
8	8	6	2

Only imaging-derived quantities and supplied segmentation masks are analyzed. No age, grade, location, or other missing clinical variable is invented.

Tumour burden versus FLAIR mean intensity



Imaging feature and tumour-burden correlations



Pairwise complete correlations among supplied labeled cases; association is not causation.

T15: held-out regression

Aim: estimate segmentation-derived tumour burden continuously.

- Task is regression, not tumour/no-tumour classification.
- Prediction-versus-reference visualisation requires regression metrics—not an invalid ROC claim.
- Local size limits confidence in generalisation.

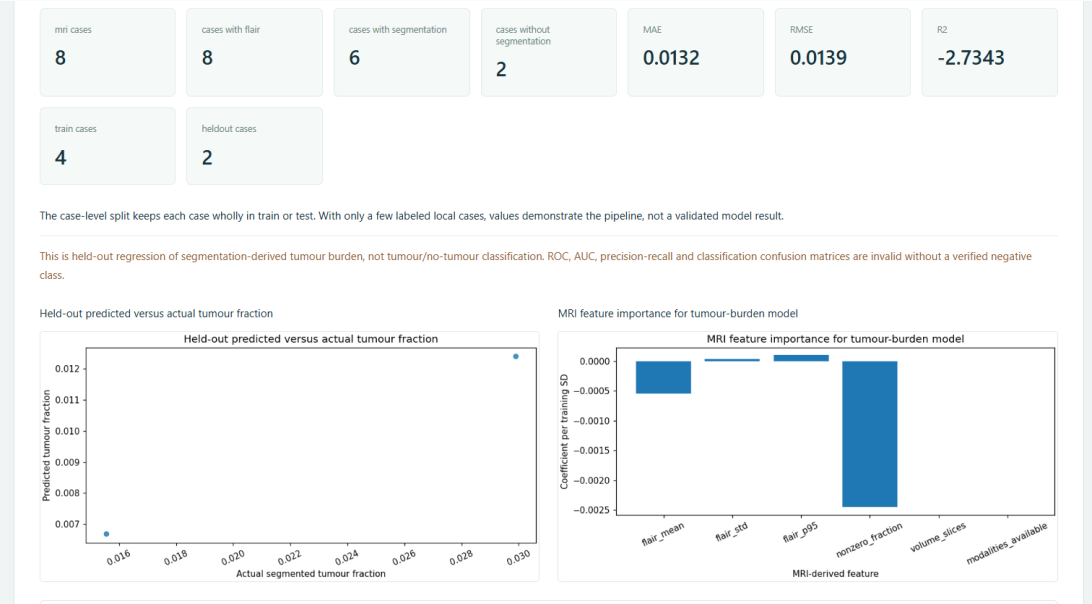
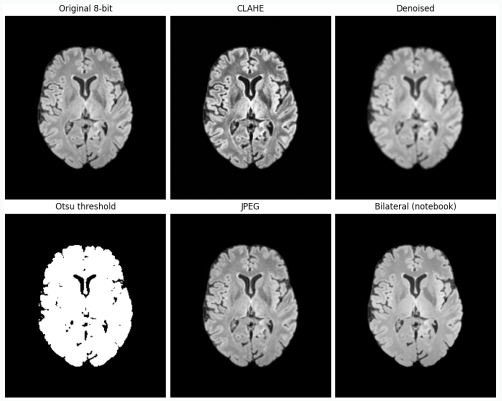
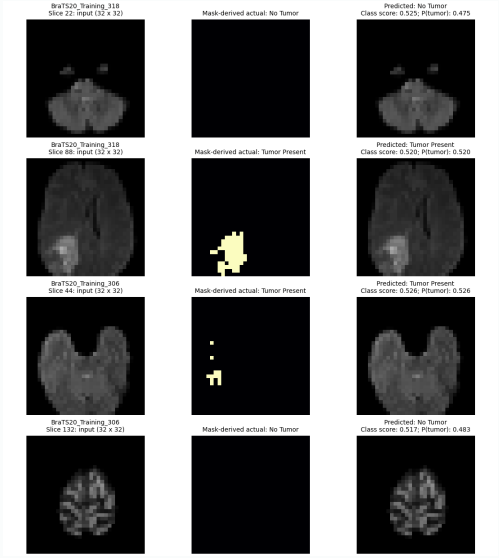


Image Analysis Studio

SIFT/RANSAC and CNN cards render actual serialized image results.

- Controls → endpoint → preprocess/model or feature algorithm → PNG/JPEG data URL → responsive cards.
- CNN cards present input, label only if available, probability/class and saved evaluation evidence.
- SIFT/RANSAC remains an independent working visual workflow.

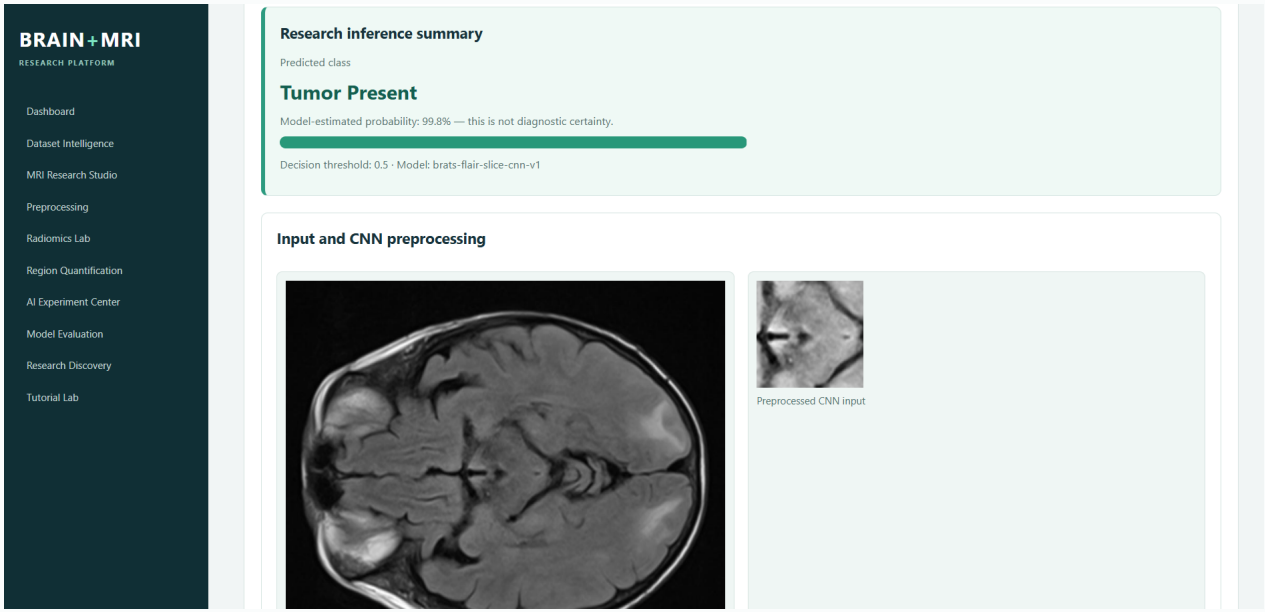




MRI Upload & AI Analysis

Persistent checkpoint inference is a separate application workflow.

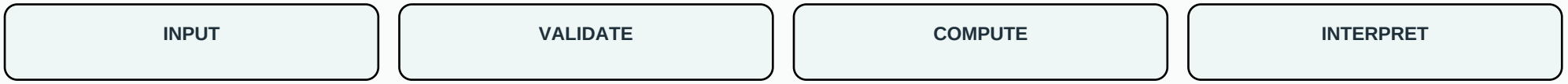
- Compatible PNG/JPEG/NIfTI FLAIR inputs are server-validated (30 MB maximum).
- L/LA/RGB/RGBA images normalise safely to one grayscale channel, then reuse the unchanged FLAIR preprocessing.
- Response includes previews, slice probabilities/classes, duration and limitations.



Research reporting and reproducibility

Reports retain actual response data and limitations.

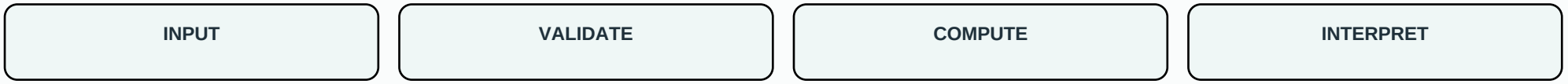
- HTML/PDF reports contain metadata, model version, preprocessing, previews and returned slice predictions.
- No dataset reference mask is presented as an upload prediction; no synthetic metrics are generated.
- Artifacts: checkpoint, metadata, history, evaluation JSON, curves, ROC and matrix.



Implementation traceability

Source files map directly to visible project behaviour.

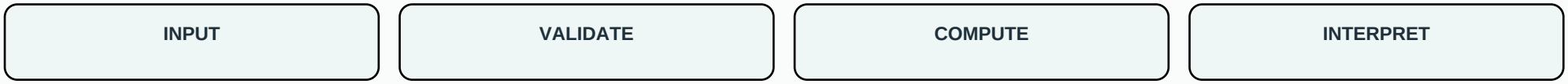
- `her_core/cnn.py`: exact preprocessing branch and saved CNN architecture.
- `scripts/train_brain_tumor_cnn.py`: patient-level experiment and persisted artifacts.
- `backend/main.py` / `backend/tutorials.py`: validated API routes; `frontend/*`: response rendering and contextual navigation.



Implementation verification

Automated checks previously completed against this project.

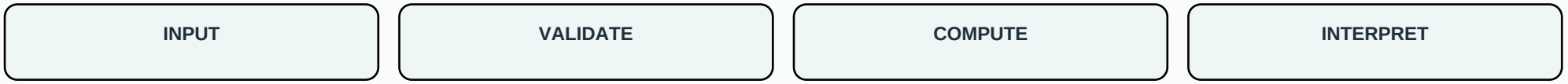
- Python: 21 passed (39 subtests) in the recorded project verification run.
- Frontend integration covered T1–T15 and CNN visualization wiring; app integration checks completed.
- This delivery validates asset presence, PPTX package slide count and all rendered PDF pages.



Limitations and safeguards

Transparency is a core technical requirement.

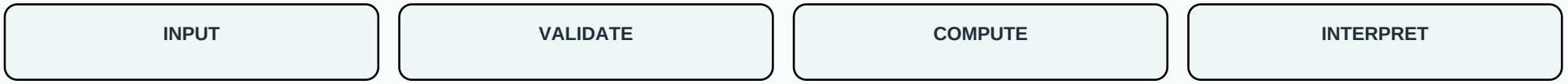
- Six labelled cases and one held-out test patient: experimental slice-level evidence only.
- No patient diagnosis, segmentation prediction, clinical confidence or arbitrary-upload accuracy is asserted.
- Raw BraTS data remains local/authorized; uploads are validated and not publicly exposed.



Future work

Substantive improvements require new data and validation.

- Multi-centre cohorts, prospective patient-level splits and independent test sets.
- Calibration/threshold selection within documented use cases and uncertainty analysis.
- Segmentation only after genuine segmentation-specific training/validation.



Conclusion

A reproducible academic platform connecting MRI engineering, tutorials and transparent AI evidence.

- T1–T15 are represented through distinct aims, methods, actual output and limitations.
- The persistent FLAIR CNN is integrated but remains experimental and slice-level.
- Source-backed visuals and metrics make the work inspectable for an academic demonstration.

